

ZY-1

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① 使用45钢. $P=4000W$. $n=750r/min$

$$P_{out} = T \cdot n / 2\pi \quad T = \frac{P}{2\pi n} = 9550 \times \frac{4}{750} = 51 N \cdot m$$

$$\alpha = 20^\circ \quad \beta = 12^\circ \quad d_1 = 300 \quad B = 90$$

$$\begin{cases} F_{t1} = \frac{2T}{d_1} = \frac{51}{300} \\ F_{r1} = F_{t1} \cdot \tan \alpha_n / \cos \beta = \frac{51 \cdot \tan 20^\circ}{\cos 12^\circ} \\ F_{a1} = F_{t1} \tan \beta = \frac{51}{300} \cdot \tan 12^\circ \\ F_{n1} = F_{t1} / \cos \alpha_t \cdot \cos \beta_b \cdot x \end{cases}$$

